

California Housing - REPORT

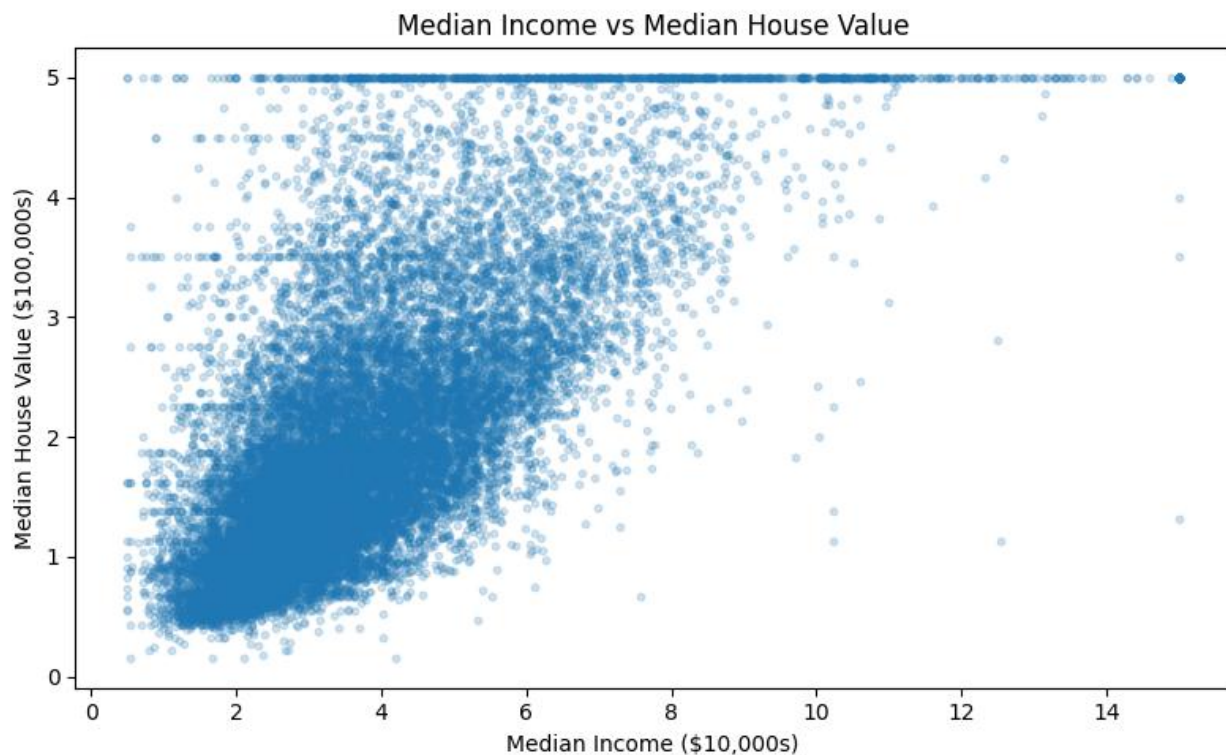
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1. Question and Data Familiarization

This study uses the California Housing dataset to examine how data structure changes the behavior of prediction methods. The analysis begins with a simple question: how strongly does median income explain median house value, and what important features of the data become visible before modeling?

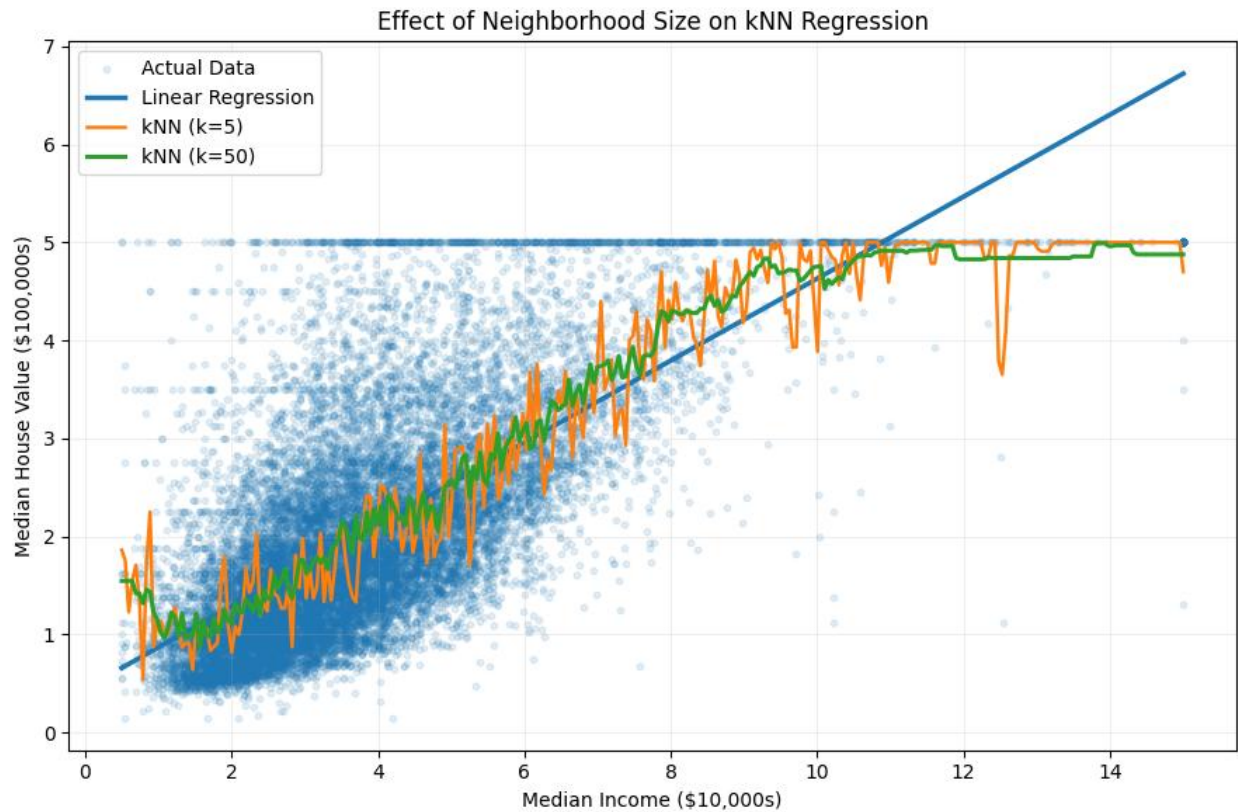
The dataset contains 20,640 districts, eight numerical input features, and median house value as the response. The response is right-skewed and has a clear ceiling near 5.0 (about \$500,000). Median income is also right-skewed. The income-value relationship is positive, but the large vertical spread shows that income alone cannot explain housing prices.

Initial insight: the target ceiling is not just a plotting detail; it constrains the response and influences what a fitted model can learn at the high-value end.



2. Global vs Local Learning

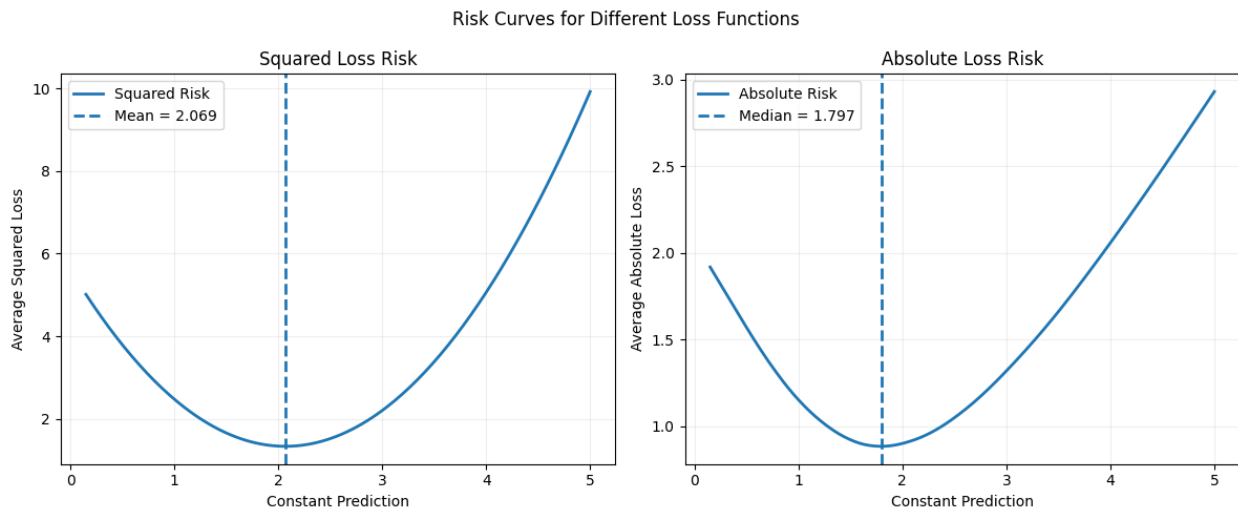
Using median income as the input, a linear regression model was compared with k-nearest-neighbour (kNN) regression. Linear regression imposes one global straight-line relationship. kNN instead predicts from nearby observations, so its shape depends strongly on the neighbourhood size.



Interpretation: small neighbourhoods provide local flexibility but are sensitive to individual observations. Increasing k sacrifices some local detail for stability. The comparison makes the bias-locality trade-off visible rather than treating model choice as only an error metric.

3. The Loss Function Changes the Best Prediction

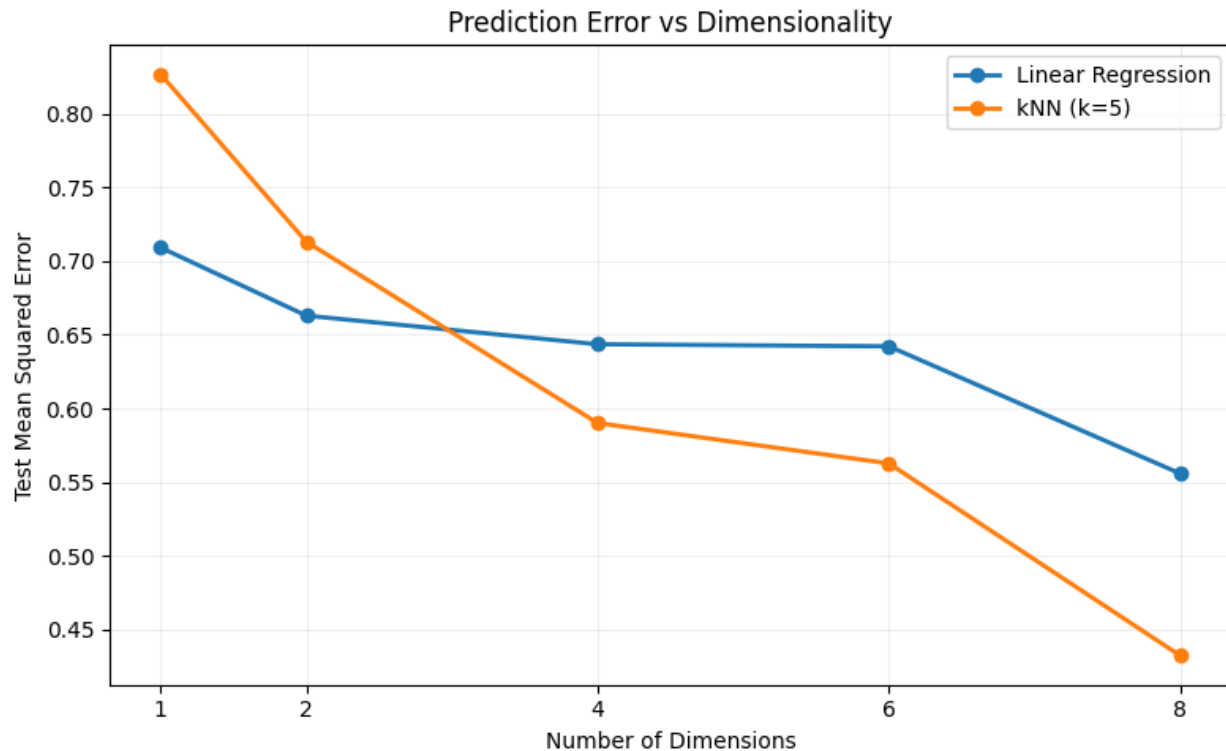
A constant predictor was evaluated under two loss functions. The sample mean house value was 2.069, while the median was 1.797. For each candidate constant, empirical squared-loss risk and absolute-loss risk were calculated.



Interpretation: the 'best' prediction is defined by the loss function. Squared loss penalizes large errors more strongly and therefore targets the mean; absolute loss is less sensitive to extreme values and targets the median. Because the response is skewed, the two optima are visibly different.

4. Dimensionality and Local Geometry

The effective input dimension was increased from 1 to 8 features. After standardization, the average nearest-neighbour distance increased from approximately 0.0002 in one dimension to 0.3435 in eight dimensions. This is the geometric effect behind the curse of dimensionality: points become less locally concentrated as dimension grows.



Important nuance: higher dimension makes nearest-neighbour geometry less local, but prediction error does not have to increase. Here, the added housing variables contain useful information, and their predictive benefit is large enough to offset the geometric disadvantage for kNN.

5. Optional Extension: Structured Linear Models

OLS was compared with Ridge and Lasso using standardized inputs. Their test MSE values were approximately 0.5559, 0.5559, and 0.5483 respectively. Ridge remained very close to OLS, while Lasso shrank coefficients more strongly and reduced the Population coefficient to approximately zero. Five-fold cross-validation favored weak regularization (best Ridge alpha about 0.1, best Lasso alpha about 0.00072). Strong Lasso regularization sharply increased error, indicating underfitting.

Model	Test MSE	Main observation
OLS	0.5559	Unregularized global linear baseline
Ridge	0.5559	Very similar coefficients and error to OLS
Lasso	0.5483	Stronger shrinkage; Population driven to approx 0

6. Conclusion

The analysis shows that model behavior is inseparable from data geometry and the objective used to measure error. Median income provides a strong global signal, but local methods reveal structure that a single straight line cannot capture. Changing the loss function changes the optimal constant prediction; increasing dimension changes what nearby means; and regularization changes how strongly a linear model relies on individual features.

The central result is that no modeling principle should be interpreted in isolation. Higher dimensionality makes local neighbourhoods geometrically weaker, yet informative features can still improve kNN accuracy. Likewise, regularization can

improve stability and sparsity, but excessive regularization can underfit. The visualizations therefore connect prediction performance to the structure of the data rather than treating the models as black boxes.

7. Story telling – Method of approach

I began by understanding the California Housing dataset before choosing any model. I checked its structure, features, missing values and summary statistics, and then used distributions and scatter plots to understand the patterns present in the data. I observed that house values were right skewed with a ceiling capped near 5.0, while median income showed a clear positive relationship with house value. Since this relationship was strong and easy to visualize, I chose median income to compare two different ways of learning: Linear Regression as a global model and kNN as a local model. I tested kNN with different neighbourhood sizes and observed that a smaller k captured more local variation but was more sensitive, while a larger k gave smoother and more stable predictions. From there, I wanted to understand what actually defines a good prediction, so I compared squared and absolute loss over different constant predictions. Using the risk curves, I verified that **squared loss was minimized near the mean**, while **absolute loss was minimized near the median**. In this way, I chose each next step based on what I observed in the previous experiment rather than treating the different parts as separate tasks.

Since kNN depends on distances between observations, I next investigated what happens when the number of features increases. I standardized the features first so that differences in their numerical scales would not unfairly influence the distance calculation. I then gradually increased the dimension from 1 to 8 and measured both nearest neighbour distance and prediction error. I found that nearest neighbour distance increased clearly with dimension, showing the geometric effect of the dimensionality. However, the kNN prediction error actually improved because the additional housing features contained useful information. Instead of forcing the expected conclusion that higher dimensions should always make kNN worse, I used the observed result to understand that useful features can sometimes compensate for the geometric disadvantage. Finally, I compared OLS, Ridge and Lasso to study how regularization changes a linear model. Lasso showed stronger coefficient shrinkage and slightly lower error, but I treated this as a comparison rather than selecting it as the final model. I then used 5 fold cross validation to test different regularization strengths and found that relatively weak regularization worked best, while excessive Lasso regularization caused underfitting. Overall, my approach was to first observe the data, decide the next question based on what I saw, test it numerically and visually, verify the result, and then draw the conclusion.